

**The Role of Digital Access and Skills in Financial Service Use in Canada:
Evidence from the Canadian Internet Use Survey – Public Use Microdata File**

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Abstract

This paper examines how digital access and digital skills are associated with the use of digital financial services (DFS) in Canada and whether these relationships vary across socio-demographic groups. Using data from the 2022 Canadian Internet Use Survey (CIUS) Public Use Microdata File, DFS use is measured across online purchasing, banking, investing, and tax filing. Linear probability models show that both digital access and digital skills are strongly associated with higher DFS use, though their importance varies by service. Digital skills increase DFS use across all four services, with more technical skills such as spreadsheet use are particularly relevant for online investing and tax filing. Smartphone access is the most important form of access, especially for routine services, while computer access is more relevant for formal financial activities. By contrast, internet quality does not have a significant independent effect once access and skills are accounted for. Despite controlling for digital capability, substantial inequalities persist: higher income and education are associated with greater DFS use, and women remain significantly less likely to engage in online investing and tax filing. Interaction results further suggest that the returns to digital skills, access, and education differ across groups, indicating that disparities in digital financial participation are not driven solely by differences in access or skills.

Introduction

As digital financial services (DFS) continue to become an essential part of modern financial practices, research on the topic has grown rapidly as financial technology (FinTech) reveals great advancements over the last few decades. These technologies have transformed how individuals' access, use, and interact with financial systems as they offer more efficient and accessible services compared to traditional banking systems (Li & Yan, 2025). FinTech innovations such as mobile banking, digital payments and online lending platforms have reduced transaction costs and expanded financial inclusion particularly for populations who have been historically excluded from the use of financial systems (Li & Yan, 2025). Despite these benefits, the digital financial gaps persist due to uneven access and use of DFS.

The digital divide was originally defined as unequal access to information and communication technologies and has evolved into a multidimensional concept that includes differences in digital skills, access, and use (Cheshmehzangi et al., 2023). Contemporary research on the topic emphasizes that inequalities appear even among those with access to technology since the meaningful use of technology depends on digital literacy and socio-economic factors (Mirazchiyski, 2025). In the context of digital financial services, the divide is particularly important as the ability to effectively use technology directly influences individuals' access to online financial tools.

The adoption and use of FinTech are shaped by multiple barriers which include limited internet access, lack of digital skills, and low financial literacy. Rural and low-income populations face constraints related infrastructure and familiarity with advanced technology which hinder their ability to adopt DFS (Wu & Peng, 2024). Similarly, certain disadvantaged

groups may experience a “dual digital divide” where both access and ability to use technology are limited, restricting their participation in economic and social activities (Barone et al., 2025). These barriers are further compounded by structural inequalities such as income, education, and geographic location which continue to shape disparities in digital use.

While DFS have the potential to promote financial inclusion, they may also worsen existing inequalities. Digital finance relies on access to electronic devices, internet connectivity, and digital literacy which tend to be unevenly distributed across populations (Yao & Ma, 2022). As a result, low-income individuals and marginalized groups typically face greater challenges in accessing and using these services, resultantly reinforcing gaps in financial participations (Yao & Ma, 2022). This has led to the phenomena known as “digital redlining” where communities with limited infrastructures and higher poverty rates experience significantly lower access to FinTech services (Friedline et al., 2020). These patterns suggest that without targeted interventions, the expansion of digital finance may not benefit all groups equally.

Conversely, there is still evidence that FinTech can help mitigate aspects of the digital divide by improving accessibility and affordability of financial services. By capitalizing mobile and digital platforms, FinTech can reduce traditional barriers such as geographic distance and high transaction costs which create opportunities for broader financial inclusion (Zhang et al., 2020). However, the extent to which these benefits are realized depends on individuals’ ability to access and effectively use digital technologies and thus demonstrating the importance of considering both access and use dimensions of the digital divide.

Despite the growth of literature on digital finance and digital divides, there remains limited research that specifically examines disparities in access to and use of DFS within high-income countries such as Canada. Much of the existing literature focuses on developing countries or rural contexts, leaving a gap in understanding how digital inequalities can manifest in more advanced economies with relatively high levels of technological infrastructure. While prior studies often examine either access to technology or adoption of FinTech, there are not many studies explicitly analyzing the interaction between access, use, and outcomes in the context of digital financial services.

This paper aims to address this gap by examining the digital divide in access to and use of digital financial services in Canada among digital users. Specifically, it examines whether differences in digital access and skills are associated with differences in digital financial service use in Canada and whether the returns to access and skills vary across demographic groups. By distinguishing between access and use, this study contributes to a more nuanced understanding of the digital divide in financial services and provides insight into the barriers that may limit the inclusive potential of digital finance. These findings have important implications for policymakers seeking to promote equitable access to financial services in an increasingly digital economy.

Data

This study uses data from the 2022 Canadian Internet Use Survey (CIUS), a nationally representative survey conducted by Statistics Canada that provides detailed information on individuals' internet usage, digital skills, access to technology, online behaviours, and demographic characteristics. This analysis is restricted to individuals who report using the internet for personal purposes in the past three months and those who have access to the internet at home, excluding mobile data plans. These restrictions reflect the relevant universe for the digital access and digital skills variables used in this study. While the CIUS survey itself is not limited to these conditions for all variables, the measures of digital access and skills require active internet use and access. As a result, the full dataset is restricted to this population, reducing the sample size from 25,118 to 21,603 observations.

The primary objective of this study is to examine how digital access and digital skills influence individuals' use of DFS in Canada among digital users while controlling for observable characteristics. Digital financial services use is defined across four key online activities: online purchasing, online banking, online investing, and online tax filing, each of which is recoded into a binary indicator for participation and non-participation. Online banking and investing are measured over the past three months, while online purchasing and tax filing are measured over the past twelve months. The exclusion of non-response and inapplicable categories (i.e., "Valid Skip", "Don't know", "Refusal", or "Not Stated") results in a negligible loss of observations, with 15 individuals excluded for online purchasing, 77 for both online banking and investing, and 4 for online tax filing. These responses are recoded as missing values (NA) and excluded from the relevant outcome-specific analyses, while remaining part of the overall sample (Table 1).

Table 1 Share of Included and Excluded Individuals for Each DFS Outcome

Inclusion and Exclusion by Digital Financial Service Outcome					
Digital financial service	Base sample	Included	Excluded	Included share (%)	Excluded share (%)
Online Purchase	21,603	21,588	15	99.931	0.069
Online Banking	21,603	21,526	77	99.644	0.356
Online Investing	21,603	21,526	77	99.644	0.356
Online Tax Filing	21,603	21,599	4	99.981	0.019

Included observations have valid binary responses (0/1); excluded observations are missing or non-response categories recoded as NA.

Digital skills are measured using four indicators of software-related activities completed in the past twelve months. These include whether the individual has: (i) copied or moved files or folders, (ii) used word processing software, (iii) created presentations or documents containing text, images, tables, or charts, and (iv) used basic functions in spreadsheet software. For each variable, responses are recoded in binary form as used or not used, with non-response and inapplicable categories coded as missing (NA) and excluded from the relevant analyses.

Digital access is measured across three dimensions: access to a smartphone, access to a desktop or laptop computer, and access to high-speed internet at home. Access to smartphones and computers is not restricted by location in the CIUS data and may include access at home, work, or in public spaces. In contrast, high-speed internet is defined as a home connection with download speeds of at least 50 megabits per second (Mbps). The high-speed internet variable is restricted to individuals who report having internet access at home and excludes mobile data connections, which may reflect a narrower view of digital access. Each access variable is recoded as a binary outcome.

Internet quality is incorporated using survey questions on the frequency of problems experienced when performing basic online activities. For basic internet activities, respondents are asked how often they experience issues when performing tasks such as browsing the web, checking email, online shopping, online banking, accessing government services, or streaming music. The response options include: “Frequent Problems”, “Occasional Problems”, and “No Problems”. Internet quality was constructed as a set of dummy variables with “No Problems” as the omitted category, while non-response categories are treated as NA. A second measure of internet quality refers to frequency of problems experienced when accessing video-based services such as video-conferencing and streaming platforms like Netflix or CraveTV. This was constructed using the same response structure and recoding as the “internet quality for basic online activities” variable. However, correlation between the two quality measures was assessed, and the video-based quality variable was excluded from the final analysis due to high multicollinearity.

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Household income is incorporated using family income quintiles which divide the sample into five groups based on total household income. Income is included in the regression as a set of categorical variables with the lowest quintile (Q1) used as the reference group. The quintiles are defined as the following:

Quintile 1 - income less than or equal to \$42,256

Quintile 2 - incomes between \$42,257 and \$72,366

Quintile 3 - incomes between \$72,367 and \$107,480

Quintile 4 - incomes between \$107,481 and \$163, 750

Quintile 5 - incomes greater than \$163, 750

The distribution of individuals across quintiles is relatively balanced, with 15.4 percent in Q1, 20.5 percent in Q2, 21.5 percent in Q3, 21.8 percent in Q4, and 20.9 percent in Q5 (Table 2).

A comprehensive set of control variables is included to account for demographic and socio-economic differences across individuals. Province of residence is grouped into five categories: Atlantic provinces (Newfoundland and Labrador, Prince Edward Island, Nova Scotia, and New Brunswick), Prairies (Manitoba, Saskatchewan, and Alberta), Quebec, British Columbia, and Ontario, with Ontario serving as the reference category. A rural dummy is constructed using population centre classifications where individuals in rural and small towns as well as those in Prince Edward Island are considered “rural” and individuals in larger urban centres are considered “urban”.

Labour market status is captured using employment status, which is treated as binary for employed and non-employed, and all other responses treated as missing (NA), with employed respondents as the omitted group. A gender dummy is also included with male as the reference group.

Age is categorized into six groups: 15-24, 25-34, 35-44, 45-54, 55-65, and 65 and above. Dummy variables are created for each age group with individuals aged 25-34 serving as the reference group.

Education is categorized into three levels: high school or less, some post-secondary (including non-degree credentials), and university degree, with university education serving as the reference group.

Immigrant status is defined as whether an individual is a landed immigrant (permanent resident) who arrived in Canada since 1952, with non-immigrants serving as the reference group. Visible minority status is similarly coded as a binary variable with non-visible minorities as the reference group. An Indigenous identity dummy was also created with non-Indigenous

individuals as the omitted category. Finally, a disability status dummy is also included with non-disabled individuals as the reference group.

Sample

All variables are constructed such that only valid responses are retained, and all non-response categories are treated as missing and excluded from the regression analysis. The final sample is restricted to individuals who reported using the internet for personal use and who have internet access which ensures that all individuals in the sample are active participants in the digital environment. As a result, the analysis focuses on variation in digital access, skills, and usage amongst internet users rather than examining disparities in basic internet access itself.

The final sample includes respondents from the 2022 Canadian Internet Use Survey who reported personal internet use in the past three months and had internet access at home, resulting in a final sample of N=21,603 observations. Table 2 reports the sample characteristics. On average, 76.9 percent of respondents reported online purchasing, 82.8 percent reported online banking, 15.9 percent reported online investing, and 31.4 percent reported online tax filing. The sample is also characterized by high levels of smartphone and high-speed internet access, at 85.9 percent and 86.7 percent respectively, though computer access is lower at 41 percent.

Digital skills are unevenly distributed across the sample, with file management (60.1 percent) and word processing (61.7 percent) more common than presentation (34.9 percent) and spreadsheet skills (45.5 percent). Table 2 further reports the distribution of the sample across income, education, gender, age, province, immigrant status, visible minority status, disability, Indigenous identity, and rural residence.

Table 2 Sample Characteristics

Variable	Mean (SD)/Percent
Percent online purchase	76.9
Percent online banking	82.8
Percent online investing	15.9
Percent online tax filing	31.3
Percent smartphone access	85.9
Percent computer access	41
Percent high-speed internet	86.7
Percent file management skills	60.1
Percent word processing skills	61.7
Percent presentation skills	34.9
Percent spreadsheet skills	45.5
Percent immigrant	18.6
Percent visible minority	19.3
Percent disability	10.4
Percent Indigenous	2.6
Percent rural	25.6
Percent internet problems (frequent)	5.7
Percent internet problems (occasional)	42.3
Percent no internet problems	52
Income quintile	
Q1	15.4
Q2	20.5
Q3	21.5
Q4	21.8
Q5	20.9

Table 2 Sample Characteristics - continued

Variable	Mean (SD)/Percent
Education	
High school or less	27.1
Some post-secondary	35.7
University	37.1
Gender	
Female	52.7
Male	47.3
Age group	
15–24	3.9
25–34	11.4
35–44	16.6
45–54	16.1
55–64	21.3
65+	30.7
Province group	
Ontario	26.2
Atlantic	17.8
Prairies	19.8
Quebec	22.2
British Columbia	14
N	21,603

Method and Empirical Specification

This study uses Ordinary Least Squares (OLS) regression to examine how digital access and digital skills are associated with individuals' use of digital financial services. OLS has also been used in related work on the digital divide and DFS gaps. Nam and Lee use OLS to examine how digital skills, literacy, and internet usage patterns explain differences in digital financial service use across income groups in South Korea (Nam & Lee, n.d.). Similarly, Aubouin uses regression-based methods to analyze how factors such as income, education, and age influence inequalities in internet access and usage, highlighting how these determinants shape different dimensions of the digital divide (Aubouin, 2023). In addition, Friedline, Naraharisetti, and Weaver examine how access to high-speed internet and socio-economic characteristics are associated with differences in FinTech usage across rural communities also while using regression analysis to capture these relationships (Friedline et al., 2020).

The analysis is carried out using separate regressions for each digital financial service outcome: online purchasing, online banking, online investing, and online tax filing. Each dependent variable is binary and captures whether the individual performed the activity within the relevant period. Estimating separate models is useful because these services do not necessarily reflect the same level or type of financial engagement. Online purchasing may capture a more routine form of digital activity, while online investing and online tax filing may reflect more advanced or formal uses of digital financial tools. Running the models separately therefore allows the relationship between access, skills, and each type of digital financial service to differ across outcomes. After estimating the separate regressions, the results are combined into one table to make it easier to compare patterns across the four services. This is also consistent with the broader literature, where different types of digital financial behaviour are often treated as distinct outcomes rather than collapsed into a single measure.

Although the dependent variables are binary, and nonlinear models such as logit or probit could in principle be used, this study uses OLS in the form of a linear probability model. One reason is that OLS coefficients are directly interpretable as changes in probability, which makes the results easier to read across multiple outcomes. More importantly, OLS is especially useful for this paper because the later analysis relies on interaction terms between key variables such as digital access, digital skills, and socio-economic characteristics. The linear structure of OLS makes these interaction terms much easier to interpret consistently across groups and outcomes. Since a major part of this paper is to examine how the relationship between access or skills and DFS use differs across groups, OLS provides a practical and interpretable framework for the analysis and serves as a useful first approximation.

The baseline specification includes measures of digital access, digital skills, and internet quality as the main explanatory variables. These are included alongside a broad set of controls capturing socio-economic and demographic characteristics. Specifically, the regressions control for income quintile, education, employment status, gender, age, province group, rural location, immigrant status, visible minority status, disability status, and Indigenous identity. These controls are included to account for observable differences that may also shape the divide in digital financial service use. The final models are estimated on the restricted sample of individuals who used the internet for personal purposes and who had internet access, which ensures consistency between the sample definition and the variables constructed in the data section.

This empirical approach also fits the broader logic of the literature on the digital divide, which increasingly treats access, skills, and outcomes as related but distinct dimensions. Rather than focusing only on whether individuals are connected, recent work emphasizes that digital inequalities depend on how effectively people can use technology and whether they are able to translate access into meaningful outcomes (Mirazchiyski, 2025). In that sense, estimating separate DFS regressions with access and skills entered jointly allows this study to examine not

only whether digital resources matter, but also how different dimensions of digital inequality are associated with different forms of financial participation.

Model

To assess how digital skills, digital access, and internet quality are associated with the use of financial services, and to examine how DFS use differs across socio-demographic groups, the baseline specification estimates separate linear probability models for each digital financial service outcome using with following estimation:

$$DFS_i = \beta_0 + \beta_1 Access_i + \beta_2 Skills_i + \beta_3 Quality_i + \beta_4 X_i + \varepsilon_i$$

In this model, DFS_i is the dependent variable and is a binary indicator equal to 1 if individual i uses a given digital financial service i.e., online purchasing, online banking, online investing, and online tax filing. The main independent variables include $Access_i$ which captures digital access through access of smartphones, desktop/ laptop computer, and high-speed internet access (50 Mbps) ; $Skills_i$ which represents digital skills measured using software-related activities such as file management, word processing, presentations, and spreadsheet functions; and $Quality_i$ which reflects internet quality based on the frequency of problems experienced during basic online activities. In addition, the model includes a vector of control variables X_i which accounts for socio-economic and demographic characteristics including province groups, and location centre, income quintile, education, employment status, gender, age group, immigrant status, visible minority status, indigenous identity, and disability status. Finally, the error term ε_i captures all unobserved factors affecting DFS use.

This model exploits variation in digital access, digital skills, internet quality, and socio-economic characteristics across individuals to examine how these factors are associated with differences in DFS use. The key independent variables (digital access and digital skills) are central to understanding how the digital divide shapes participation in DFS.

Digital access variables capture whether individuals have the necessary infrastructure to engage with digital finance. These variables reflect the foundational layer of the digital divide, as without reliable access to devices, individuals may be unable to participate in online financial activities at all. Digital skills on the other hand capture individuals' ability to effectively use digital technologies. Even amongst individuals with similar levels of access, differences in skills may lead to substantial differences in how technology is used. In the context of digital financial services, activities such as online investing or tax filing often require higher levels of digital proficiency compared to more routine activities like online purchasing or banking. As a result,

the inclusion of multiple skills measures allows the analysis to capture whether individuals with stronger digital capabilities are more likely to engage in these services.

Internet quality is included to complement the access variable. It accounts for differences in the reliability of digital access. Even when individuals have access to devices and internet connection, frequent connectivity issues may discourage or limit their use of DFS. By including a measure of internet quality, the model accounts for this additional dimension of access that may affect usage outcomes.

The vector of control variables captures the broader socio-economic and demographic factors that may also influence DFS use. For example, income and education may affect both access to technology and the ability to use it effectively, while age may capture generational differences in digital familiarity. Geographic variables such as location centre and province groups account for differences in infrastructure and availability of services. Identity-related variables including immigrant status, visible minority status, Indigenous identity, and disability status are included to capture structural inequalities that may shape both access to and use of digital technologies. Controlling for these factors allows the analysis to better isolate the relationship between digital access and digital skills on the use of digital financial services.

Given the use of a linear probability model, the estimated coefficients can be interpreted as the change in probability of using a given digital financial service associated with a one-unit change in the corresponding independent variable, holding all other factors constant.

Results

Baseline Differences in Digital Financial Service Use

To assess how digital skills, digital access, and internet quality are associated with digital financial service use, Table 3 presents a baseline model that includes measures of digital skills, digital access, internet quality, income, education, geography, labour force status, gender, and demographic characteristics. The coefficients in Table 3 show average differences in the probability of using online purchasing, online banking, online investing, and online tax filing, conditional on the included controls. The table contains two broad sets of coefficients of interest. First, the coefficients on digital skills and digital access indicate how different dimensions of digital capability are associated with the use of digital financial services. Second, the socio-economic and demographic coefficients show how digital financial service use differs across groups once those capability measures are taken into account.

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Table 3 OLS Estimates of Digital Financial Services Use

	Online Purchase	Online Banking	Online Investing	Online Tax Filing
File management skills	0.141*** (0.009)	0.095*** (0.008)	0.039*** (0.007)	0.056*** (0.010)
Word processing skills	0.112*** (0.010)	0.089*** (0.009)	0.017* (0.008)	0.060*** (0.011)
Presentation skills	0.012* (0.006)	-0.008 (0.005)	0.046*** (0.008)	0.008 (0.010)
Spreadsheet skills	0.029*** (0.007)	0.044*** (0.006)	0.084*** (0.008)	0.101*** (0.010)
Smartphone access	0.174*** (0.012)	0.195*** (0.012)	0.031*** (0.008)	0.023* (0.011)
Computer access	0.007 (0.006)	0.015** (0.005)	0.016** (0.006)	0.023** (0.007)
High-speed internet	0.036*** (0.009)	0.038*** (0.009)	0.003 (0.008)	0.011 (0.010)
Internet quality: occasional problems	0.015** (0.006)	0.001 (0.005)	-0.008 (0.006)	-0.009 (0.007)
Internet quality: frequent problems	-0.006 (0.014)	0.004 (0.012)	-0.011 (0.012)	-0.016 (0.016)
Income Q2	0.013 (0.011)	0.018+ (0.010)	0.001 (0.008)	0.012 (0.012)
Income Q3	0.049*** (0.011)	0.027** (0.010)	0.017+ (0.009)	0.013 (0.012)
Income Q4	0.054*** (0.010)	0.026** (0.010)	0.030** (0.009)	0.035** (0.012)
Income Q5	0.051*** (0.010)	0.030** (0.010)	0.077*** (0.010)	-0.002 (0.013)
High school or less	-0.079*** (0.009)	-0.042*** (0.008)	-0.076*** (0.008)	-0.053*** (0.010)
Some post-secondary	-0.021*** (0.006)	-0.008 (0.006)	-0.052*** (0.007)	-0.024** (0.009)
Rural	0.010 (0.007)	0.002 (0.007)	-0.020** (0.007)	-0.062*** (0.009)
Atlantic	-0.023* (0.009)	0.011 (0.009)	-0.042*** (0.009)	0.070*** (0.012)
Prairies	-0.011	0.014+	-0.021*	0.040***

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Linear probability models with HC1 robust standard errors.

Reference groups: No internet problems, Income Q1, University, Ontario, Urban, Employed, Male, Non-immigrant, Not a visible minority, No disability, Non-Indigenous, and Age 25–34.

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Table 3 OLS Estimates of Digital Financial Services Use - Continued

	Online Purchase	Online Banking	Online Investing	Online Tax Filing
	(0.008)	(0.008)	(0.009)	(0.011)
Quebec	-0.038***	0.035***	-0.044***	-0.061***
	(0.008)	(0.007)	(0.008)	(0.010)
British Columbia	0.004	0.017*	0.019+	0.072***
	(0.009)	(0.008)	(0.010)	(0.012)
Not employed	-0.019*	-0.040***	0.001	-0.001
	(0.008)	(0.007)	(0.007)	(0.009)
Female	-0.005	0.009+	-0.120***	-0.054***
	(0.006)	(0.005)	(0.006)	(0.007)
Immigrant	-0.028**	-0.007	-0.019*	-0.013
	(0.009)	(0.008)	(0.009)	(0.011)
Visible minority	-0.049***	-0.054***	0.012	0.009
	(0.009)	(0.008)	(0.010)	(0.012)
Disability	-0.001	0.001	-0.011	0.013
	(0.010)	(0.009)	(0.009)	(0.012)
Indigenous	-0.044*	-0.000	-0.012	0.034
	(0.019)	(0.017)	(0.017)	(0.023)
Age 15–24	-0.013	-0.102***	-0.070***	-0.202***
	(0.016)	(0.016)	(0.017)	(0.019)
Age 35–44	0.018*	-0.005	-0.034**	-0.006
	(0.008)	(0.007)	(0.011)	(0.014)
Age 45–54	0.005	-0.004	-0.052***	-0.022
	(0.009)	(0.008)	(0.012)	(0.014)
Age 55–64	-0.023*	0.007	-0.048***	-0.028*
	(0.009)	(0.008)	(0.011)	(0.014)
Age 65+	-0.060***	0.004	-0.012	-0.048***
	(0.011)	(0.009)	(0.012)	(0.014)
Num.Obs.	16615	16579	16579	16619
R2	0.220	0.170	0.121	0.082
R2 Adj.	0.219	0.169	0.120	0.080

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Linear probability models with HC1 robust standard errors.

Reference groups: No internet problems, Income Q1, University, Ontario, Urban, Employed, Male, Non-immigrant, Not a visible minority, No disability, Non-Indigenous, and Age 25–34.

Estimates for digital skills show a clear and consistent relationship between digital capability and digital financial service use. File management skills and word processing skills are positive in all four specifications, with the largest coefficients appearing for online purchasing and online banking. File management, for example, ranges from 0.039 for online investing to 0.141 for online purchasing, while word processing ranges from a smaller but still significant 0.017 for online investing to 0.112 for online purchasing. Spreadsheet skills are also positive across all four outcomes, but stand out especially for online investing (0.084) and online tax filing (0.101), suggesting that more structured or numerically oriented skills are particularly relevant for the more formal digital financial services. Presentation skills are much less consistent. They are only weakly positive for online purchasing (0.012) and clearly positive for online investing (0.046), but are not statistically significant for online banking or online tax filing. Taken together, these estimates suggest that digital skills matter across the board, but their importance increases as the service becomes more information-intensive or administratively complex.

The access coefficients show a similarly strong, though somewhat more differentiated, pattern. Smartphone access is the strongest and most consistent measure of digital access in the table. It is positively associated with all four outcomes, with particularly large coefficients for online purchasing (0.174) and online banking (0.195), and smaller but still significant coefficients for online investing (0.031) and online tax filing (0.023). Computer access is more modest. It is not significantly associated with online purchasing, but is positively related to online banking (0.015), online investing (0.016), and online tax filing (0.023). High-speed internet is positively associated with online purchasing (0.036) and online banking (0.038), but is not significant for investing or tax filing. These estimates suggest that different forms of access matter for different services. Smartphone access appears especially important for routine digital financial activities, while computer access is more relevant for tasks that are more document-heavy or formally structured.

By contrast, internet quality does not emerge as an important independent predictor in any of the four outcomes. Once the model accounts for access, skills, and socio-economic characteristics, the coefficient on internet quality is small and insignificant throughout. This suggests that within a sample already restricted to internet users, differences in connection quality are less central than differences in the type of access and the ability to use digital tools effectively.

The socio-economic controls show that digital financial service use remains stratified even after controlling for digital capability. Income exhibits a clear gradient for online purchasing, online banking, and especially online investing. Compared with the lowest income quintile, respondents in the higher quintiles are generally more likely to use these services, with the strongest coefficient appearing for income quintile 5 in online investing (0.077). Tax filing shows a weaker and less consistent income gradient, although income quintile 4 remains

positively associated with filing online (0.035). Education is also strongly related to DFS use. Relative to respondents with a university degree, those with high school or less are less likely to use every service, with coefficients ranging from -0.042 for online banking to -0.079 for online purchasing. The penalties are also substantial for online investing (-0.076) and online tax filing (-0.053). Respondents with some post-secondary education also show negative coefficients for several outcomes, particularly online investing (-0.052) and tax filing (-0.024). These estimates suggest that the educational gradient remains strong even after controlling for digital access and digital skills, indicating that education captures more than technological exposure alone.

The geography variables also reveal meaningful differences across place. Rural residence is not significantly associated with online purchasing or banking, but is negatively associated with online investing (-0.020) and especially online tax filing (-0.062). Provincial differences are also notable. Relative to Ontario, respondents in the Atlantic provinces are less likely to purchase online (-0.023) and invest online (-0.042), but more likely to file taxes online (0.070). The Prairies show mixed results across outcomes, with no significant effect for online purchasing, a small positive association for online banking (0.014), a negative and statistically significant effect for online investing (-0.021), and a positive and statistically significant effect for online tax filing (0.040). Quebec shows the broadest negative pattern, with lower online purchasing (-0.038), lower online investing (-0.044), and lower online tax filing (-0.061), though higher online banking use (0.035). British Columbia, by contrast, shows a strong positive coefficient for online tax filing (0.072) and a smaller positive coefficient for online banking (0.017). These regional patterns suggest that digital financial participation is not uniform across Canada and may reflect institutional, service, or behavioural differences across provinces.

Labour market status matters mainly for the more routine services. Individuals who are not employed are less likely to purchase online (-0.019) and less likely to use online banking (-0.040), while there are no meaningful differences for investing or tax filing. This suggests that labour force attachment may be more closely tied to day-to-day transactional needs than to the use of more specialized digital financial platforms.

The demographic coefficients show further heterogeneity in digital financial service use. Gender differences are especially pronounced in the more formal services. There is no significant difference between women and men in online purchasing, and only a weakly positive difference in online banking, but women are substantially less likely to use online investing (-0.120) and online tax filing (-0.054). The coefficient on immigrant status is smaller, but still indicates lower use of online purchasing (-0.028) and online investing (-0.019). Visible minority respondents are less likely to purchase online (-0.049) and less likely to use online banking (-0.054), though there are no significant differences for investing or tax filing. Disability is not significantly associated with any of the four outcomes, while Indigenous identity is only negatively associated with online purchasing (-0.044).

Finally, the age coefficients suggest that digital financial participation follows a life-cycle pattern rather than a simple linear trend. Relative to respondents aged 25 to 34, those aged 15 to 24 are much less likely to use online banking (-0.102), online investing (-0.070), and especially online tax filing (-0.202), though they are not significantly different in online purchasing. The middle age groups show smaller and more selective differences, particularly lower investing among those aged 35 to 44 (-0.034) and 45 to 54 (-0.052). Older respondents also appear disadvantaged in several dimensions. Those aged 55 to 64 are less likely to purchase online (-0.023), less likely to invest online (-0.048), and less likely to file taxes online (-0.028), while those aged 65 and above are less likely to purchase online (-0.060) and file taxes online (-0.048). These findings suggest that younger and older respondents face different forms of exclusion: younger respondents appear less engaged in formal financial services, while older respondents are more disadvantaged in some every day and administrative forms of digital participation.

Heterogeneity in Digital Financial Service Use Across Groups

To highlight the heterogeneity not captured by the baseline OLS estimates, the analysis extends the main specification by introducing interaction terms between key dimensions of the digital divide and selected socio-demographic characteristics. Because internet quality did not emerge as a significant predictor in the baseline OLS models, it is not carried forward into the interaction analysis. The focus instead shifts to the two dimensions that appeared most relevant in the baseline results: digital access and digital skills, along with education, which remained strongly associated with digital financial service use across outcomes. These variables are interacted with gender, visible minority status, and age, where age is recoded into a binary comparison between individuals under 55 and those aged 55 and above in order to capture broader generational differences in digital financial behaviour more clearly. These interaction models allow the analysis to move beyond average effects and assess whether the relationship between digital capability and digital financial service use differs systematically across groups. The coefficients of interest in these models are, first, the baseline group indicators, which measure the gap relative to the omitted reference category, and second, the interaction terms, which show whether the returns to access, skills, or education differ across groups.

Visible Minority x Digital Access

The results in Table 4 suggest that visible minorities generally use digital financial services slightly less than non-visible minorities, although the gap is not large and is only clearly significant for online banking. In the restructured version of the table, the natural reference group is non-visible minorities without access. Relative to that group, the coefficient on visible minority status without access is small and not significant for online purchasing, investing, and tax filing, and only clearly negative for online banking. Smartphone access remains strongly positive for both groups in routine services: for non-visible minorities, the coefficients are 0.177

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for online purchasing and 0.189 for online banking, while for visible minorities they remain sizable at 0.154 and 0.170, respectively. This is why the gap appears only slightly smaller once smartphone access is added. High-speed internet does not show the same equalizing pattern and is even weakly negative for visible minorities in banking (-0.067). The most notable exception is tax filing, where computer access for visible minorities is relatively large (0.105), suggesting that computers matter more than general connectivity for this more administrative digital service. Overall, these estimates support the interpretation that visible minorities use some services slightly less than non-visible minorities, but that access itself does not appear to be the main source of the gap.

Table 4 Visible Minority x Digital Access

Visible Minority x Digital Access: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Non-visible minority without access (reference)	0.000	0.000	0.000	0.000
with smartphone access	0.177*** (0.013)	0.189*** (0.012)	0.021** (0.008)	0.029* (0.012)
with computer access	0.003 (0.006)	0.013* (0.005)	0.014* (0.006)	0.016+ (0.008)
with high speed internet access	0.044*** (0.010)	0.046*** (0.009)	0.001 (0.009)	0.014 (0.011)
Visible minority without access	-0.053 (0.041)	-0.065 (0.043)	-0.033 (0.032)	0.055 (0.042)
with smartphone access	0.154*** (0.027)	0.170*** (0.025)	0.026 (0.025)	0.024 (0.030)
with computer access	-0.034 (0.042)	-0.038 (0.043)	-0.009 (0.032)	0.105* (0.043)
with high speed internet access	-0.041 (0.038)	-0.067+ (0.040)	-0.028 (0.027)	0.057 (0.039)
R ²	0.219	0.168	0.119	0.076
Adjusted R ²	0.218	0.166	0.117	0.075

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Non visible minorities are the reference group. All regressions control for digital skills, income, education, gender, age, geography, labour status, internet quality and demographic characteristics.

Visible Minority x Digital Skills

A somewhat stronger pattern appears in the interaction between digital skills and visible minority status estimates in Table 5. Visible minorities without skills are already less likely than non-visible minorities without skills to purchase online (-0.074) and use online banking (-0.110), which suggests that part of the visible minority gap is already present before additional digital capability is introduced. At the same time, the returns to specific skills are uneven. Among non-visible minorities, file management and spreadsheet skills are strongly positive across several outcomes, especially investing and tax filing. For visible minorities, the strongest gains come from spreadsheet skills, which are positive for online investing (0.065) and especially online tax filing (0.138). By contrast, presentation skills are sharply negative for online banking among visible minorities (-0.127). These estimates support the idea that the visible minority gap is not

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simply about lacking skills altogether, but also about which skills are most useful for translating digital capability into actual financial participation.

Table 5 Visible Minority x Digital Skills

Visible Minority x Digital Skills: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Non-visible minority without skills (reference)	0.000	0.000	0.000	0.000
with file management skills	0.146*** (0.010)	0.100*** (0.008)	0.038*** (0.008)	0.061*** (0.011)
with word processing skills	0.112*** (0.011)	0.074*** (0.009)	0.013 (0.008)	0.058*** (0.012)
with presentation skills	0.004 (0.007)	-0.015** (0.006)	0.040*** (0.009)	-0.006 (0.011)
with spreadsheet skills	0.019* (0.008)	0.036*** (0.007)	0.083*** (0.008)	0.098*** (0.011)
Visible minority without skills	-0.074*** (0.018)	-0.110*** (0.018)	-0.020 (0.013)	0.006 (0.018)
with file management skills	0.053* (0.022)	-0.027 (0.022)	0.025 (0.019)	0.043+ (0.024)
with word processing skills	0.031 (0.023)	0.017 (0.022)	0.016 (0.019)	0.043+ (0.025)
with presentation skills	-0.040+ (0.023)	-0.127*** (0.022)	0.035 (0.021)	0.015 (0.027)
with spreadsheet skills	-0.006 (0.024)	-0.029 (0.023)	0.065** (0.021)	0.138*** (0.028)
R ²	0.220	0.169	0.119	0.076
Adjusted R ²	0.219	0.168	0.117	0.075

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Non visible minorities are the reference group. All regressions control for digital access, income, education, gender, age, geography, labour status, internet quality and demographic characteristics.

Age x Digital Access

Table 6 shows the results of interactions between access and individuals ages 55 and above. In general, older individuals are less likely to purchase online and somewhat less likely to use online banking, but are somewhat surprisingly more likely to invest online. The strongest age disadvantage appears in online purchasing, where older adults without access begin from a lower baseline (-0.151). Smartphone and high-speed access help narrow that gap. Smartphone access among those under 55 is already strongly positive for purchasing (0.124) and especially banking (0.185), but for older adults access also matters. In online investing, the coefficients for older adults with smartphones (0.092), computers (0.083), and high-speed internet (0.065) are all positive. This is why the age pattern looks different for investing than for purchasing. These estimates may suggest that older individuals are already more likely to invest online, and that the additional benefit of smartphone access is smaller for them than for younger respondents. So, while access clearly matters for purchasing and banking among older adults, it appears less central for explaining differences in investing.

Table 6 Age x Digital Access

Age 55+ x Digital Access: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Under age 55 without access (reference)	0.000	0.000	0.000	0.000
with smartphone access	0.124*** (0.030)	0.185*** (0.031)	0.065** (0.020)	0.027 (0.030)
with computer access	0.001 (0.007)	0.012* (0.006)	0.022* (0.009)	0.016 (0.010)
with high-speed internet	0.011 (0.012)	0.010 (0.011)	0.017 (0.014)	0.018 (0.017)
Age 55+ without access	-0.151*** (0.035)	-0.036 (0.036)	0.073** (0.025)	-0.002 (0.036)
with smartphone access	0.038 (0.034)	0.158*** (0.035)	0.092*** (0.025)	0.021 (0.035)
with computer access	-0.138*** (0.035)	-0.016 (0.036)	0.083** (0.025)	0.028 (0.036)
with high-speed internet	-0.093** (0.034)	0.021 (0.035)	0.065** (0.024)	0.005 (0.035)
R ²	0.220	0.168	0.119	0.076
Adjusted R ²	0.218	0.166	0.117	0.074

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Individuals under age 55 are the reference group. All regressions control for digital skills, income, education, gender, geography, labour status, internet quality and demographic characteristics.

Age x Skills

The results in Table 7 suggest that older people without skills do not necessarily use online banking and investing less than younger people without skills. The clearest positive interaction terms for older respondents are for file management skills, particularly in online banking (0.111) and online investing (0.059). Spreadsheet skills are also clearly important, especially for online investing (0.090), and older individuals with spreadsheet skills appear broadly similar to younger individuals with spreadsheet skills in online banking, tax filing, and investing. At the same time, the absence of skills among older adults is associated with lower online purchasing (-0.117) and lower tax filing (-0.055). Not all skills work positively: presentation skills are negative for tax filing among older adults (-0.093), and spreadsheet skills are negative for online purchasing (-0.097). These estimates support the interpretation that for older individuals, file-based and task-oriented skills matter most, while more general or less practical forms of digital familiarity do not necessarily translate into higher use of financial digital services.

Age x Education

The interaction between education and age results in Table 8 show a more cumulative disadvantage, though the interactions also suggest that some of the age-based gaps shrink among the more educated groups. Individuals under 55 with high school or less are already much less likely than younger university-educated respondents to use all four services, especially online

Table 7 Age x Skills

Age 55+ x Digital Skills: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Under age 55 without skills (reference)	0.000	0.000	0.000	0.000
with file management skills	0.092*** (0.011)	0.066*** (0.011)	0.035** (0.011)	0.024+ (0.015)
with word processing skills	0.096*** (0.013)	0.088*** (0.013)	0.003 (0.012)	0.019 (0.016)
with presentation skills	0.009 (0.007)	-0.010 (0.007)	0.075*** (0.010)	0.029* (0.013)
with spreadsheet skills	0.041*** (0.009)	0.049*** (0.009)	0.092*** (0.010)	0.114*** (0.014)
Age 55+ without skills	-0.117*** (0.015)	-0.009 (0.014)	0.015+ (0.008)	-0.055*** (0.013)
with file management skills	0.069*** (0.017)	0.111*** (0.015)	0.059*** (0.012)	0.030+ (0.018)
with word processing skills	-0.001 (0.018)	0.071*** (0.016)	0.046*** (0.012)	0.023 (0.018)
with presentation skills	-0.089*** (0.017)	-0.024 (0.015)	0.013 (0.015)	-0.093*** (0.020)
with spreadsheet skills	-0.097*** (0.018)	0.035* (0.016)	0.090*** (0.014)	0.042* (0.019)
R ²	0.223	0.168	0.121	0.078
Adjusted R ²	0.221	0.167	0.119	0.076

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Individuals under age 55 are the reference group. All regressions control for digital access, income, education, gender, geography, labour status, internet quality and demographic characteristics.

investing (-0.112) and tax filing (-0.114). Among older adults, the penalties remain substantial: those aged 55 and above with high school or less are less likely to purchase online (-0.127), invest online (-0.077), and file taxes online (-0.084). Even older adults with university education remain less likely than younger university-educated respondents to purchase online (-0.025) and file taxes online (-0.045). These estimates suggest that the differences between older and younger individuals are relatively small among the more educated groups, but much larger among those with lower education.

Gender x Access

The estimates in Table 9 suggest that women are generally less likely to use digital financial services than men, but that access does not consistently reduce this gap. Smartphone access is strongly positive for both men and women in routine services such as purchasing and banking: for men the coefficients are 0.165 and 0.181, while for women they are 0.135 and 0.165. However, this pattern does not carry over to the more formal services. Smartphone access is negative for women in online investing (-0.035), and women without access are already less likely than men without access to purchase online (-0.060) and file taxes online (-0.072). Computer access is also less beneficial for women, with negative coefficients for purchasing (-0.065) and tax filing (-0.058). These are the main coefficients supporting the conclusion that

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Table 8 Age x Education

Age 55+ x Education: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Under age 55 with university education (reference)	0.000	0.000	0.000	0.000
with high school or less	-0.057*** (0.011)	-0.060*** (0.010)	-0.112*** (0.011)	-0.114*** (0.014)
with some post-secondary	-0.013+ (0.007)	-0.018** (0.007)	-0.065*** (0.010)	-0.051*** (0.012)
Age 55+ with university education	-0.025** (0.008)	0.009 (0.007)	-0.019+ (0.011)	-0.045*** (0.013)
with high school or less	-0.127*** (0.012)	-0.039*** (0.010)	-0.077*** (0.011)	-0.084*** (0.014)
with some post-secondary	-0.056*** (0.009)	0.006 (0.008)	-0.064*** (0.010)	-0.047*** (0.013)
R ²	0.220	0.167	0.119	0.077
Adjusted R ²	0.218	0.166	0.118	0.075

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Individuals under age 55 with university education are the reference group. All regressions control for digital skills, digital access, income, geography, labour status, internet quality, gender and demographic characteristics.

access helps women in routine services, but does not eliminate and may even reinforce gender gaps in more formal digital financial services.

Table 9 Gender x Access

Gender x Digital Access: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Male without access (reference)	0.000	0.000	0.000	0.000
with smartphone access	0.165*** (0.016)	0.181*** (0.016)	0.064*** (0.012)	0.022 (0.016)
with computer access	0.017* (0.008)	0.015* (0.007)	0.031*** (0.009)	0.031** (0.010)
with high-speed internet	0.017 (0.012)	0.023* (0.011)	0.012 (0.013)	-0.002 (0.015)
Female without access	-0.060* (0.025)	-0.042 (0.026)	-0.022 (0.017)	-0.072** (0.024)
with smartphone access	0.135*** (0.022)	0.165*** (0.022)	-0.035* (0.018)	-0.048* (0.023)
with computer access	-0.065* (0.025)	-0.026 (0.026)	-0.022 (0.018)	-0.058* (0.025)
with high-speed internet	0.003 (0.024)	0.013 (0.024)	-0.029+ (0.017)	-0.045+ (0.023)
R ²	0.220	0.168	0.120	0.076
Adjusted R ²	0.218	0.166	0.119	0.075

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Males are the reference group. All regressions control for digital skills, income, education, age, geography, labour status, internet quality and demographic characteristics.

Gender x Education

A much clearer pattern emerges in the Gender x Education results in Table 10. Relative to men with university education, women with the same level of education appear equally likely to use online purchasing and banking, but remain less likely to use online investing (-0.171) and online tax filing (-0.082). For men, education is strongly and positively associated with all four services. Compared to men with university education, men with some post-secondary are less likely to invest online (-0.089) and less likely to file taxes online (-0.046), while men with high school or less are even less likely to invest online (-0.141) and file taxes online (-0.114). For women, education also improves purchasing and banking, but the pattern changes sharply for investing. Women with university education are already less likely to invest online (-0.171), and this gap widens further for those with some post-secondary (-0.196) and high school or less (-0.199). A similar pattern is observed for tax filing, with women with university education less likely to file taxes online (-0.082), and larger gaps for those with some post-secondary (-0.097) and high school or less (-0.122).

Table 10 Gender x Education

Gender x Education: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Male with university education (reference)	0.000	0.000	0.000	0.000
with some post-secondary	-0.024** (0.009)	-0.020* (0.008)	-0.089*** (0.011)	-0.046*** (0.013)
with high school or less	-0.083*** (0.011)	-0.074*** (0.010)	-0.141*** (0.011)	-0.114*** (0.013)
Female with university education	-0.008 (0.007)	-0.006 (0.006)	-0.171*** (0.010)	-0.082*** (0.012)
with some post-secondary	-0.025** (0.009)	-0.009 (0.008)	-0.196*** (0.010)	-0.097*** (0.012)
with high school or less	-0.084*** (0.011)	-0.040*** (0.010)	-0.199*** (0.010)	-0.122*** (0.013)
R ²	0.219	0.168	0.122	0.077
Adjusted R ²	0.218	0.166	0.121	0.075

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Males with university education are the reference group. All regressions control for digital skills, digital access, income, age, geography, labour status, internet quality and demographic characteristics.

Gender x Digital Skills

The results in Table 11 reinforce the conclusions for gender-based digital financial services use. For men, digital skills are broadly positive across outcomes, with especially strong effects of spreadsheet skills for online investing (0.115) and online tax filing (0.098). For women, however, the pattern is narrower. Women without skills are already less likely to invest online (-0.040) and file taxes online (-0.035). File management and word processing remain strongly positive for women in purchasing and banking, but these same skills do not produce comparable gains in investing. Word processing is actually negative for women's online investing (-0.038), and presentation skills are negative for women's online tax filing (-0.056).

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Only spreadsheet skills remain clearly positive for women in online banking (0.062) and online tax filing (0.075).

Table 11 Gender x Digital Skills

Gender x Digital Skills: Group Effects				
Group	Online Purchase	Online Banking	Online Investing	Online Tax Filing
Male without skills (reference)	0.000	0.000	0.000	0.000
with file management skills	0.153*** (0.012)	0.100*** (0.011)	0.048*** (0.012)	0.065*** (0.015)
with word processing skills	0.090*** (0.013)	0.085*** (0.012)	0.032** (0.012)	0.053*** (0.016)
with presentation skills	0.027** (0.008)	-0.014+ (0.007)	0.066*** (0.012)	0.014 (0.014)
with spreadsheet skills	0.028** (0.010)	0.050*** (0.009)	0.115*** (0.013)	0.098*** (0.015)
Female without skills	-0.007 (0.013)	0.020 (0.013)	-0.040*** (0.007)	-0.035** (0.011)
with file management skills	0.125*** (0.016)	0.110*** (0.015)	-0.006 (0.010)	0.013 (0.016)
with word processing skills	0.124*** (0.016)	0.106*** (0.014)	-0.038*** (0.010)	0.019 (0.017)
with presentation skills	-0.010 (0.016)	0.006 (0.015)	-0.021+ (0.012)	-0.056** (0.017)
with spreadsheet skills	0.025 (0.016)	0.062*** (0.015)	0.011 (0.011)	0.075*** (0.017)
R ²	0.219	0.167	0.125	0.076
Adjusted R ²	0.218	0.166	0.124	0.075

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Males are the reference group. All regressions control for digital access, income, education, age, geography, labour status, internet quality and demographic characteristics.

These results suggest that both education and digital skills increase digital financial service use, but their returns are weaker for women, especially in the more formal services such as online investing and tax filing. In other words, gender differences in DFS use persist even after accounting for differences in education and digital skills.

Conclusion

This paper examines how digital access and digital skills are associated with the use of digital financial services in Canada among internet users. The results show that both access and skills play an important role in shaping digital financial participation, though their effects differ across services. Digital skills are consistently associated with higher use across all four outcomes, with more structured skills such as spreadsheet use being particularly important for more formal activities such as online investing and online tax filing. Digital access also matters, but in different ways. Smartphone access is most relevant for routine services such as online purchasing

and online banking, while computer access is more important for more formal financial activities.

At the same time, substantial inequalities remain even after accounting for digital capability. Higher-income individuals, particularly those in the upper income quintiles, are more likely to use digital financial services, especially online investing. Individuals with lower levels of education, particularly those with high school or less, are consistently less likely to use all four services compared to those with a university degree. Gender differences are also concentrated in more formal services, with women less likely than men to use online investing and online tax filing even when they have similar levels of access and skills. Geographic differences persist as well, with individuals living in rural areas and in provinces such as Quebec generally showing lower participation in several digital financial services relative to those in Ontario.

The interaction results further show that the relationship between digital capability and digital financial service use differs across groups. Visible minorities, for example, do not appear to benefit from all forms of access in the same way as non-visible minorities, suggesting that access alone does not fully explain differences in use. Older individuals, particularly those aged 55 and above, rely more heavily on structured and task-oriented digital skills, while younger individuals, especially those under 25, are less engaged in more formal financial services such as investing and tax filing. Gender differences also persist within these interactions, as women remain less likely than men to use more formal digital financial services even when they have similar levels of education and digital skills.

Despite these findings, several limitations should be noted. First, this analysis does not address the digital divide more broadly, as the sample is restricted to individuals who already use the internet and have access at home. As a result, the paper focuses on differences among digitally connected individuals rather than inequalities in basic access. Second, while the linear probability model provides a practical and interpretable framework, a nonlinear model such as logit or probit may be more appropriate given the high share of users for some services. Third, the measures of digital access and digital skills are limited by the available survey data. Access is captured through whether individuals report having access to devices, without distinguishing between personal, shared, or public access, and skills are based on self-reported activities rather than direct measures of ability. Finally, the age results suggest that younger and older individuals engage with digital financial services in fundamentally different ways. Younger individuals are less involved in formal financial services, while older individuals face different barriers related to adoption and use, indicating that these groups may require separate approaches to understanding digital financial behaviour.

Overall, the findings suggest that improving access to technology alone is not sufficient to ensure equitable participation in digital financial services. Differences in digital skills, socio-economic characteristics, and the ability to translate access into meaningful use continue to shape digital financial inclusion in Canada.

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